

The Magnetotelluric method: Examples from uppermost mantle to the uppermost crust

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Summary

The natural source Magnetotelluric (MT) is becoming an attractive electromagnetic method to explore deep tectonic settings as well as very shallow targets in mineral and geothermal explorations. Thanks to recent advances in instrumentation and software developments, the MT is a standard method in many geophysical applications. In our contribution we will give a quick introduction on the method and show examples from around the world to explore structures from the uppermost mantle to near surface mineral and geothermal applications.

Introduction

It has been always believed that the MT method is mainly used to study deep tectonic structures and is less sensitive towards shallow structures for economic purposes, e.g., mineral, and geothermal exploration. This belief is based on the old generation of MT instruments which measure very low frequency data. However, the recent developments in MT instrumentation as well as in processing software and modelling codes have changed the perception on the MT method. For example, the new generations of MT instruments allow recording a wide spectra of MT data covering frequencies from $\sim 10,000$ Hz to $\sim 10 \times 10^{-4}$ Hz with one single set of instruments and low signal-to-noise ratio. Moreover, the new developments in processing software includes, among others, various filtering to process the recorded time series and provide high quality MT data.

Beside the developments in instrumentation and data processing software, inversion codes made the MT methods even more attractive in providing a complete view on the electrical conductive distribution of the subsurface. For instant, the rapid developments in High Performance Computation (HPC) technology as well as developing new mathematical approaches to solve large systems of equations helped modelling MT data considerably. For example, 3D modelling of MT data is becoming a MUST in many large and small surveys.

Adding to all of these, the low logistic associated with the field work making the MT method an attractive and cost-effective method when compared to other geophysical method which require a man-made source of energy to inject in the subsurface.

In this contribution we will discuss MT results on a wide range of scales: From continental scale arrays to explore the deep tectonic structures of the Earth to exceedingly small and local arrays for economic purposes.

Method

The MT method relies on naturally occurring EM field variations generated in the ionosphere and by lightning discharges, which induce currents into the Earth. The theory behind the method has been described before, for example, Cagniard (1953), Keller (1988) and Weaver (1994). Field data are acquired by measuring variations of electric (**E**) and magnetic (**B**) fields at Earth's surface, typically using non-polarizing electrodes and induction coil magnetometers. The magnetic field components B_x, B_y and B_z are oriented in N-S, E-W and vertical directions, and the two horizontal electric dipoles E_x and E_y are oriented in N-S and E-W directions. From the recorded time-series of the horizontal electric and magnetic fields, the so-called impedance tensor (**Z**) is calculated. Mathematically, impedance tensor (**Z**) is described as the ratio of the electric to the magnetic field and written as:

$$\mathbf{E} = \mathbf{Z}\mathbf{B} \rightarrow \mathbf{Z} = \mathbf{E}\mathbf{B}^{-1}$$

The impedance tensor depends on frequency (or period) and is usually presented as apparent resistivity and phase:

$$\left\{ \begin{array}{l} \rho_a^{i,j} = 0.2T|Z_{i,j}| \\ \phi_{i,j} = \arctan\left(\frac{\text{Im}(Z_{i,j})}{\text{Re}(Z_{i,j})}\right) \end{array} \right\} \quad i, j = x, y$$

The apparent resistive and phase values derived from the MT data carry information on the electrical conductivity of the subsurface at various depths based on the recorded frequency: High frequency data senses the shallow part whereas low frequencies cover deep structures.

Examples

Our main contribution is to show the effectiveness of MT method in covering very deep tectonic structures and near surface economic features. Hence, we will show and discuss examples covering a wide scale using arrays from around the World.

Starting from the uppermost mantle structures we will discuss results from long period (10 - 20,000 s) MT data acquired across the continental USA on a quasi-regular grid of ~ 70 km spacing as an electromagnetic component of the National Science Foundation Earth Scope/USArray Program. The 3D inversion results from the semi-continental USArray array reveal the deep (up to ~ 350 km depth) tectonic structures of the North American tectonic

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plate and the close-by oceanic Gorda and Juan de Fuca Plates (Meqbel et al., 2014). The West-East section at a latitude of 42.5°N extract from the 3D model which is obtained from the USArray data summarizes most of the tectonic regimes in NW Pacific. Namely, at the western part of the section the 3D inversion results show the subduction system between the oceanic Gorda Plate and the North American Plate. In the central part of the section, the 3D model exposes the extensional regime in the Northern Basin and Range area. At the easternmost part of the section the 3D model started to sense the cratonic stable part North American tectonic Plate (Figure 1).

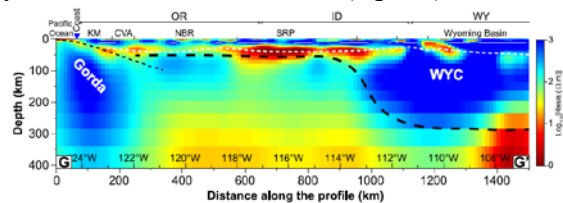


Figure 1: Representative east-west cross section (latitude 42.5°N). Several tectonic regimes can be observed on this section and will be discussed extensively in our contribution. OR: Oregon; ID: Idaho; WY: Wyoming; KM: Klamath Mountains; CVA: Cascade Volcanic Arc; NBR: Northern Basin and Range; SRP: Snake River Plain; WYC: Wyoming Craton.

On continental scale arrays, we will discuss results obtained from a) South African MT Experiment (SAMTEX; e.g., Khoza et al., 2013), b) SINOPROBE in China (Dong et al., 2014) and c) Australian Lithospheric Architecture Magnetotelluric Project (AusLAMP; Robertson et al. 2016; Robertson et al., 2020; Thiel et al., 2020; Kirkby et al., 2020).

On crustal structures there are numerous publications using 2D and/or 3D inversion of MT data, however, we will focus on 2D, and 3D inversion results of MT data collected in the framework of DESIRE-Project: DEAd Sea Integrated REsearch, which aims at studying the crustal structures of the Dead Sea Basin using various geophysical results. The main finding from the MT data is summarized in Figure 2 (Meqbel et al., 2016)

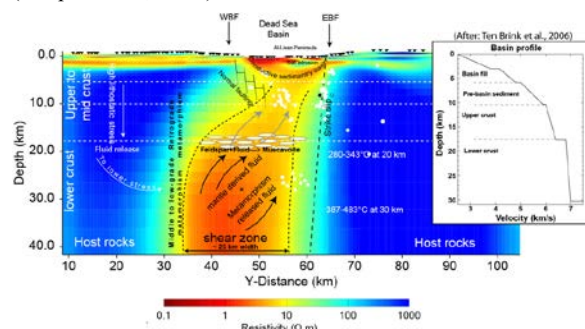


Figure 2: Summary of the 2D conductivity model obtained from MT data across the Dead Sea Basin superimposed by results from

other geophysical and geodynamic methods within the DESIRE-Project.

The most interesting crustal feature observed in Figure 2 is a conductive anomaly which we interpreted as a share zone associated with the Dead Sea Transform (DST) fault which strikes the Dead Sea basin. DST has its footprint in opening the crust at this place and it is well imaged in the 2D electrical conductivity model.

On the uppermost crust and for economic purposes there are many publications and commercial projects which show the value of MT method, particularly when integrated with other methods. For instance, the application of MT in geothermal explorations is becoming a MUST in many surveys. This is because MT data are extremely sensitive to the alteration cap (conductive rocks) which is associated with geothermal system. Examples of using MT in geothermal explorations can be found in projects from around the World. Here we focus on few examples from the western part of Turkey. For examples, in a commercial project, MT data were recorded at 350 sites to understand the geological structures of northern margin of Gediz Graben in western Turkey and to determine the locations of the production and re-injection boreholes. 3D inversion results have been discussed in Erdogan and Candansayar (2014). Figure 3 shows two cross sections from the final 3D electrical conductivity model of using 350 MT sites. The most prominent feature in these sections is the conductive layer which appears to perfectly match the Clay-Sandstone layer and interpreted as the alteration cap of the geothermal system under consideration.

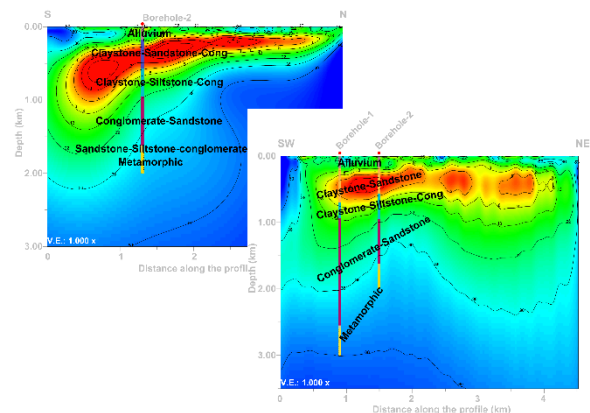


Figure 3: Two cross sections extracted from a 3D conductive model of inverting 350 MT sites northern margin of Gediz Graben in western Turkey. The results show a perfect match between the lithological columns from previously drilled boreholes and the conductive anomalies.

The conventional MT method is either surface-based on Land or on sea floor in marine EM. For new advances in natural sources EM method, we will report and show

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examples on Z TEM method; an Airborne EM method which is based on recording the time variations of the vertical magnetic component in the air and process these against land based two horizontal magnetic components. Collecting and processing the ZTEM in that way results into a dataset which is similar to the traditional tipper data. However, the advantage of ZTEM when compared to surface-based tipper is that data acquisition is much faster and one can collect data on a much dense array. Figure 4 shows an example from a 3D conductive model obtained by inverting a very dense ZTEM array from Mt. Milligan, North-Central British Columbia mining area in Canada.

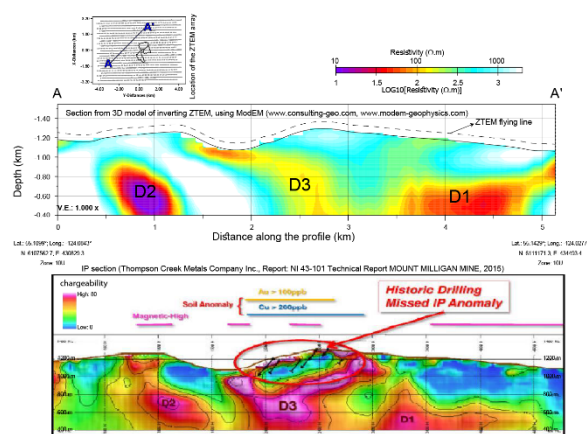


Figure 4: Upper panel: A section from the 3D electrical conductivity model derived from a very dense ZTEM array. The lower panel shows the chargeability obtained from IP method along the same section. The small inset at the top shows the ZTEM array and the location of the extracted section.

Conclusion

The Magnetotelluric method has been advanced in the last 20 years in terms of instrumentations as well as in processing, modelling and interpretations software. The advances allowed the application of MT method on various scales to delineate very deep tectonic structures and very shallow feature for economic usages. The examples we discussed and other examples from around the World show the added values of MT method if and only if the data are collected, processed and modelled in a proper way

References

Cagniard L., 1953, Basic theory of the magneto-telluric method of geophysical prospecting, *Geophysics*, vol. 18, Pages 605-635.

Dong, H., Wei, W., Ye, G., Jin, S., Jones, A. G., Jing, J., Zhang, L., Xie, C., Zhang, F., and Wang, H., 2014, Three-dimensional electrical structure of the crust and upper mantle in Ordos Block and adjacent area: Evidence of regional lithospheric modification, *Geochem. Geophys. Geosyst.*, 15, 2414–2425, doi:10.1002/2014GC005270.

Erdoğan E. and Candansayar M. Emin, 2017, The conductivity structure of the Gediz Graben geothermal area extracted from 2D and 3D magnetotelluric inversion: Synthetic and field data applications. *Geothermics*, Volume 65, Pages 170-179, doi:10.1016/j.geothermics.2016.09.007.

Khoza, T. D., A. G. Jones, M. R. Muller, R. L. Evans, M. P. Miensoopust, and S. J. Webb, 2013, Lithospheric structure of an Archean craton and adjacent mobile belt revealed from 2-D and 3-D inversion of magnetotelluric data: Example from southern Congo craton in northern Namibia, *J. Geophys. Res. Solid Earth*, 118, doi:10.1002/jgrb.50258.

Keller, V. G. 1988, *Physical Laws, Electromagnetic Induction and Rock Properties*. Chap. 2, pages 13–51 of: Nabighian, Misac N. (ed), *Electromagnetic Methods in Applied Geophysics*. Society of Exploration Geophysics (SEG).

Kirkby, A., Musgrave, R.J., Czarnota, K., Doublier, M.P., Duan, J., Cayley, R.A., Kyi, D., 2020. Lithospheric architecture of a Phanerozoic orogen from magnetotellurics: AusLAMP in the Tasmanides, southeast Australia. *Tectonophysics*, v. 793, 228560.

Robertson K., Heinsohn G. and Thiel S., 2016, Lithospheric reworking at the Proterozoic–Phanerozoic transition of Australia imaged using AusLAMP Magnetotelluric data, *Earth and Planetary Science Letters*, Volume 452, Pages 27–35.

Robertson, K., Thiel, S. and Meqbel, N., 2020 Quality over quantity: on workflow and model space exploration of 3D inversion of MT data. *Earth Planets Space* 72, 2. doi.org/10.1186/s40623-019-1125-4.

Meqbel, N., G. D. Egbert, P. E. Wannamaker, A. Kelbert, and A. Schultz, 2014, Deep electrical resistivity structure of the northwestern U.S. derived from 3-D inversion of USArray magnetotelluric data: *Earth and Planetary Science Letters*, 402, Pages 290–304.

Meqbel N., Weckmann U., Muñoz G., Ritter O., 2016, Crustal metamorphic fluid flux beneath the Dead Sea Basin: constraints from 2-D and 3-D magnetotelluric modelling, *Geophysical Journal International*, Volume 207, Issue 3, Pages 1609–1629, doi.org/10.1093/gji/ggw359

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Thiel, S., Goleby, B.R., Pawley, M.J. and Heinson G., 2020, AusLAMP 3D MT imaging of an intracontinental deformation zone, Musgrave Province, Central Australia. *Earth Planets Space* 72, 98. doi.org/10.1186/s40623-020-01223-0.

Weaver J.T., *Mathematical Methods for Geoelectromagnetic Induction*, 1994, Taunton, Somerset, England Applied and Engineering Mathematic Series, Research Studies Press